

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED
COMPUTING 2019, ICRTAC 2019

Automatic Determination of K in Distributed K-Means Clustering

Dinesh Kumar Kotary^a, Satyasai Jagannath Nanda^a^aMalaviya National Institute of Technology, Jaipur-302017, India

Abstract

Traditional K-Means based distributed data clustering require number of clusters as input which is difficult to obtain in case of a real life application like wireless sensor network. To mitigate this issue here an Automatic Distributed K-Means (ADK-Means) algorithm is proposed. In this algorithm cluster assignment is carried out with point symmetry based distance instead of Euclidean distance, to effectively detect arbitrary shaped clusters. The performance of proposed algorithm is demonstrated on four synthetic and one practical dataset. Three cluster quality evaluation methods have been employed and comparative analysis is done with existing distributed K-Means. The improved performance is up to 12.6% in terms of Silhouette index reported as compared to existing approach.

© 2019 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019.

Keywords: Point symmetry distance; Wireless sensor networks; Dunn index.

1. Introduction

Wireless sensor networks (WSN) consist of tiny nodes which have sensing and communication capability. In the last decade WSN are deployed for applications like monitoring of hardly accessible environments, detection of enemy army in battlefield, finding location of forest fire etc [2]. These applications require processing a large volume of data. Previously centralized clustering is used to process the data which has large communication burden and less security of individual node. Distributed data clustering reduces communication burden and keeps security of individual sensor nodes.

Distributed data clustering (DDC) is applicable where data is located at several sensor nodes and there is need of finding hidden pattern in data. In DDC every sensor node finds local clustering result and summarized result is shared

*Corresponding Author. Tel.: +919882712104

Email Address: 2015rec9510@mnit.ac.in

with neighbouring nodes. All neighbour nodes do clustering with help of their local data and received summary from other nodes. After convergence global clustering result is obtained at each node and we can access the clustering pattern of whole area by accessing the information of single node [1]. The practical application of DDC is clustering of computer records (text, images, zip files) located at different laptops. Clustering of books at different libraries and clustering of criminal records at several police stations etc [4].

Distributed data clustering started when Bandyopadhyay et al. in 2006 proposed distributed clustering for data residing at different stations [5]. This algorithm faces difficulty in synchronization. Datta et al. in [8] developed a distributed K-Means clustering for dynamic networks (there is a change in the neighbourhood. This method has limitation when data is not uniformly distributed over the network. Forero et al. [9] solved the distributed clustering problem as an distributed optimization. The convergence parameter in this algorithm make unstable when not selected properly. Pedrycz et al. [11] developed distributed fuzzy clustering in which any data point can be associated with multiple clusters. In literature [3] distributed clustering with gossip-based sharing is used for static and dynamic networks. This algorithm is not much effective to detect arbitrary shape clusters. Most of the above discussed techniques require number of clusters as input which is rather difficult in real life scenarios.

This paper proposes automatic distributed clustering using K-Means approach. The algorithm selects number of clusters automatically using leap based approach. Further for detecting arbitrary shape clusters the algorithm perform clustering using point symmetry distance instead of Euclidean distance. The proposed technique is validated on four synthetic and one real life datasets.

The rest of the article is organized as: Section 2 defines basic problem formulation of distributed clustering in WSN. The proposed automatic distributed clustering is described in section 3. Section 4 gives detailed description of dataset used and parameter settings. The simulation result and discussion is reported in section 5. The concluding remark about proposed algorithm is discussed in section 6.

2. Distributed clustering problem in Wireless sensor network

Consider a WSN consists of K number of sensors, $\forall k \in [1, K]$. Network is fully connected and every node has a set of $(K-1)$ neighbours. Every node has a dataset $W_k = [w_k^1, w_k^2, \dots, w_k^n, \dots, w_k^N]$, $W_k \in R^d$ where d is dimension and N is number of data samples recorded by node k . The data point of any k^{th} node is denoted by w_k^n , $\forall n \in [1, N]$. In clustering, each sensor dataset W_k gets partitioned into Z clusters C_k^z , $\forall z \in [1, Z]$ with corresponding cluster centre denoted by c_k^z . The value of Z is determined by algorithm itself and not user dependent. In this paper distributed clustering is performed based on partitioned approach which has the following properties:

1. The local dataset at every sensor node comprise sum of data points present in all the clusters: $W_k = \bigcup_{z \in Z} C_k^z$.
2. At every sensor node, each cluster C_k^z should possess at least one data point C_k^z , $\forall z \in [1, Z]$, $\forall k \in [1, K]$.
3. At every sensor node, there should not be any common data points between two different clusters:
 $C_k^{z_1} \cap C_k^{z_2} = \emptyset$, $\forall z_1 \neq z_2$ and $\forall z_1, z_2 \in [1, Z]$
4. After convergence of clusters using diffusion mode, at all peers of WSN the cluster center of same clusters should be nearly same $C_{k_1}^z \cap C_{k_2}^z = \emptyset$, $\forall k_1, k_2 \in [1, K]$ and $\forall z \in [1, Z]$.

In distributed clustering the intra-cluster distance between data element w_k^n and cluster center c_k^z is minimized.

$$\text{Minimize } \zeta = \sum_{k \in K} \sum_{z \in Z} \sum_{n \in C_k^z} \|w_k^n - c_k^z\|^2 \quad (1)$$

where $n \in [1, n_i]$, and n_i is the number of data points present in cluster C_k^z .

3. Proposed Automatic distributed clustering using K-Means

3.1 Last leap method for selecting number of clusters

It has been observed that the separation between the cluster centers show significant leaps with increase in cluster number. This concept is used in the proposed method to detect the optimum number of clusters in the sensor dataset. Each sensor node finds separation between cluster centers which shows many leaps in the separation pattern as in Fig.1 after that the last significant leap is identified and the cluster number corresponding to last leap is termed as number of cluster at particular node. Since there are k nodes then the mutual sharing method is obtained for final number of clusters. The separation between cluster centers is obtained from:

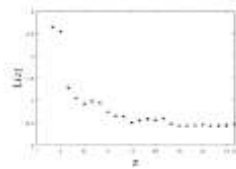


Fig.1: Showing leaps for three clusters (Last major leap at cluster index 3)

$$G_k^z = \min \|c_k^i - c_k^j\| \quad \text{where } i, j = 2, 3 \dots Z_{\max} \text{ and } i \neq j \quad (2)$$

where c_k^i represent the i^{th} cluster of node k and $Z_{\max}$ is maximum number of clusters. $Z_{\max}$ is taken as square root of total data points at a sensor. The leaps in the separation are given by:

$$L_k(z) = \frac{G_k^z - G_k^{z+1}}{G_k^z} \quad (3)$$

The number of clusters at each node is identified by the cluster index which shows maximum leap in $L_k(z)$

$$Z_k = \arg \max L_k(z) \quad (4)$$

The cluster number obtained from each node is shared with neighboring nodes and average number of cluster is identified. The integer value of average number of cluster is considered as number of clusters (Z) at each node.

3.2 Distributed clustering using K-Means by sharing of cluster centers

The step by step procedure of the algorithm is outlined as:

Step 1: Find out Number of clusters using leap based method.

Step 2: Every sensor node randomly select Z cluster centers from the dataset. Initialize Lagrangian multiplier for each sensor and choose value of $\mu > 0$

Step 3: Assign each data point to cluster center from which it has least Euclidean distance.

Step 4: Share the cluster centers to all neighboring sensor nodes and Lagrangian multiplier to next immediate node.

Step 5: Sensor k updates its cluster center by the following:

$$c_k^z = \frac{\sum_{n \in C_k^z} (w_k^n) + \sum_{k \in K} (2\mu [c_k^z - \lambda_k^z + \lambda_{k1}^z])}{2\mu [W] + |C_k^z|} \quad (5)$$

Step 6: Peer $k \in K$ update the value of its Lagrange multipliers $\lambda_{k,k1}^z, \forall z \in Z, k1 \in K$ as:

$$\lambda_{k,k1}^z = \lambda_{k,k1}^z + \mu (c_k^z - c_k^z) \quad (6)$$

Repeat from step 3 to 6 till convergence condition occurs.

Step 7: Final cluster assignment: Now a point w_k^i joins a cluster z based on the condition below:

$$\begin{aligned} &\text{If } (d_{ps}(w_k^i, c_k^z)) / (d_e(w_k^i, c_k^z)) \leq \lambda \\ &\text{and } d_{ps}(w_k^i, c_k^z) \leq d_{ps}(w_k^i, c_k^y), \forall y \in Z, z \neq y \end{aligned}$$

Now a point w_k^i joins a cluster s based on the condition below:

$$\begin{aligned} &\text{If } (d_{ps}(w_k^i, c_k^z)) / (d_e(w_k^i, c_k^z)) > \lambda \\ &\text{and } d_e(w_k^i, c_k^s) \leq d_e(w_k^i, c_k^y), \forall y \in Z, z \neq s \end{aligned}$$

Where $d_e(w_k^i, c_k^z)$ defines Euclidean distance between data item w_k^i to cluster center c_k^z and $d_{ps}(w_k^i, c_k^z)$ is point symmetry distance and can be calculated as [12].

4. Simulation Setup

The simulation studies of proposed ADK-Means and DK-Means [9] algorithms are carried out in MATLAB R2015a in an Intel Core i7 processor with 8 GB RAM and 1 TB hard disk on Windows 10 (64-bit) platform. Both algorithms are allowed to run for 100 iterations, and the obtained performance is recorded. Four number of sensors are considered for all five experiments. The value of μ is taken as 0.5 for each algorithm. Three performance evaluation indices are used for accessing the quality of distributed clustering as follows.

4.1 Minkowski Score

Minkowski Score (MS) is a crucial criterion for evaluating cluster quality subject to the condition when the true partition is known apriori. MS score describes the amount of incorrectly clustered data items. In wireless sensor

network for distributed clustering the Minkowski Score is given by:

$$MS_{dc} = \left(\frac{1}{K} \right) \sum_{k=1}^K MS_k \quad (7)$$

Where MS_{dc} is average MS of all peer nodes and MS_k is MS of k^{th} peer node and can be calculated as in [10].

4.2 Silhouette index

The Silhouette index (SI) [10] is one of the popularly used performance indices in cluster analysis. A higher value of SI signifies that data points have better matching to its cluster and less similarity from other clusters. Silhouette index for distributed clustering is defined as:

$$SI_{dc} = \left(\frac{1}{NK} \right) \sum_{k=1}^K \sum_{n=1}^N SI(w_k^n) \quad (8)$$

where N is number of data points in a sensor and K is number of sensor. The Silhouette at any k^{th} sensor associated with n^{th} data is calculated as in [10].

4.3 Dunn index

Dunn index [10] is another performance index to validate cluster quality. It is defined as the ratio of minimum inter-cluster distance to maximum cluster size. The high value of Dunn index leads to better clustering. In WSN for distributed clustering the Dunn index is given by:

$$DI_{dc} = \left(\frac{1}{K} \right) \sum_{k=1}^K DI(Z_k) \quad (9)$$

where Dunn index for one sensor can be calculated as in [10].

5. Results and Discussions

5.1. Synthetic dataset results

Four synthetic dataset namely as Mixed_3_2, Mixed_5_2, AD_5_2, and AD_4_3 shown in Fig.2 used for validation of proposed algorithm. The same dataset is used for all four sensor nodes and performance parameters are recorded. The leap base method is used to determine the number of clusters and corresponding leap pattern is reported in Fig.3. It is clearly showing dataset Mixed_3_2 has last major leap at cluster three, Mixed_5_2 at five, AD_5_2 at five and AD_4_3 has at four cluster. In Table 1 the maximum clusters taken and final number of clusters obtained is reported. Dataset Mixed_3_2 is used in [13] having two dimensional 577 elements. Distributed clustering results of all four sensors are reported in Fig. 4 and Fig. 5 for ADK-means and DK-Means respectively. Most of the points are correctly clustered using ADK-Means while DK-means clustering shows poor performance. The Minkowski score of ADK-Means is less which shows improved performance as compared to DK-Means. The Dunn index is better for ADK-Means as compared to DK-Means.

Dataset Mixed_5_2 has 850 points having two dimensions. The clustering results are reported in Fig. 6 and Fig. 7 using ADK-Means and DK-Means respectively. The Minkowski score and Dunn index is better in ADK-Means for this dataset.

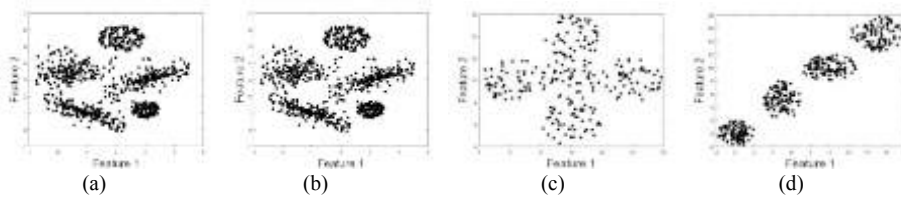


Fig. 2: Datasets given to different sensors: (a) Mixed_3_2, (b) Mixed_5_2, (c) AD_5_2, (d) AD_4_3

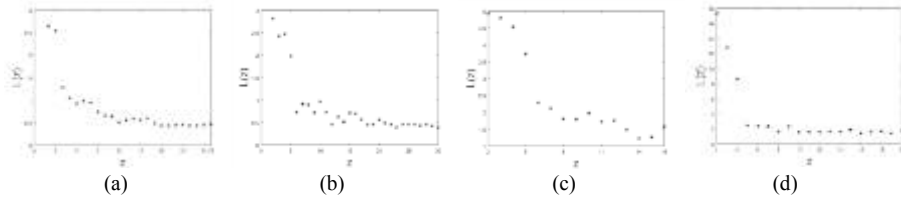


Fig. 3: Leap patterns for different dataset: (a) Mixed_3_2, (b) Mixed_5_2, (c) AD_5_2, (d) AD_4_3

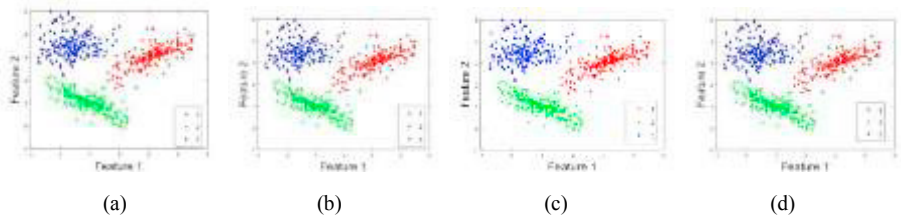


Fig. 4: Clustering results for Mixed_4_3 dataset using ADK-Means technique at different nodes : (a) node 1, (b) node 2, (c) node 3, (d) node 4

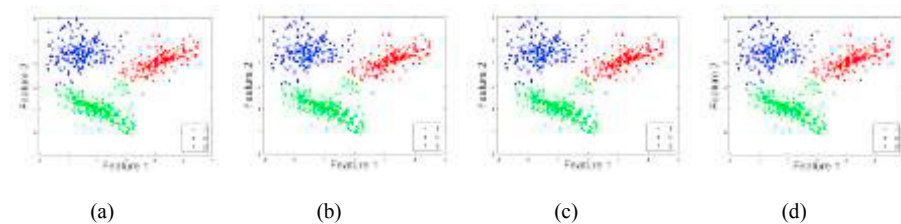


Fig. 5: Clustering results for Mixed_3_2 dataset using DK-Means technique at different nodes : (a) node 1, (b) node 2, (c) node 3, (d) node 4

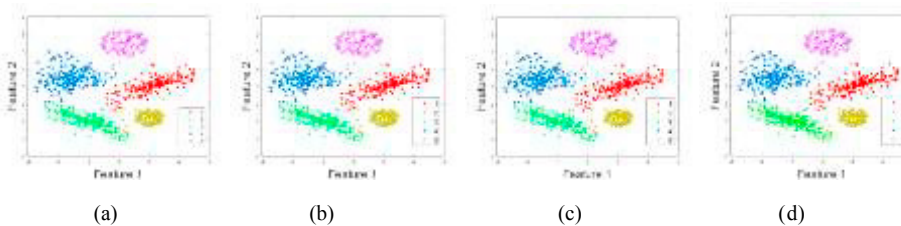


Fig. 6: Clustering results for Mixed_5_2 dataset using ADK-Means technique at different nodes : (a) node 1, (b) node 2, (c) node 3, (d) node 4

Dataset AD_5_2 is taken from [6] which has two dimension and 250 points. It consists of five clusters each having 50 points. Clustering result using ADK-Means and DK-Means are obtained and results are shown in Table 1.

Dataset AD_4_3 has three dimension and three perfectly separable cluster. The clustering result using proposed method and existing DK-Means approach is described in Fig.8 and Fig.9. It is observed that both the algorithm obtain perfect clustering for this dataset.

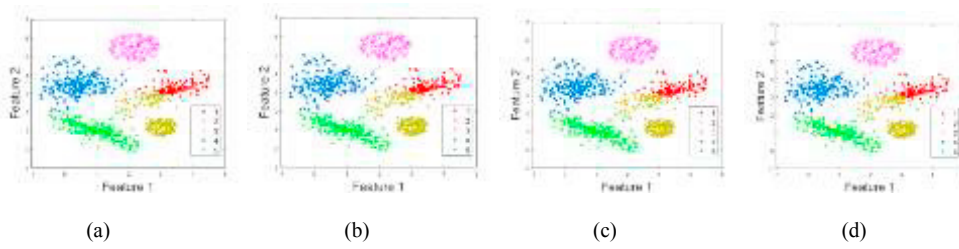


Fig. 7: Clustering results for Mixed_5_2 dataset using DK-Means technique at different nodes : (a) node 1, (b) node 2, (c) node 3, (d) node 4

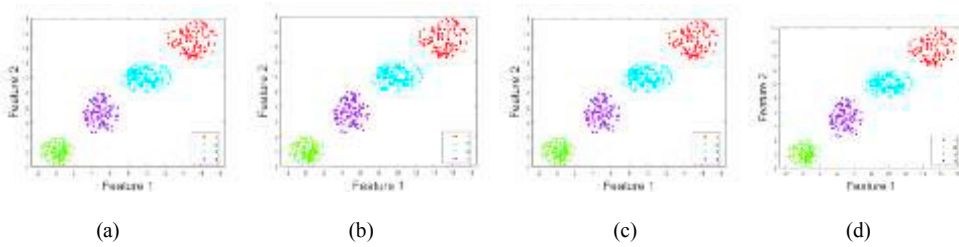


Fig. 8: Clustering results for AD_4_3 dataset using ADK-Means technique at different nodes : (a) node 1, (b) node 2, (c) node 3, (d) node 4

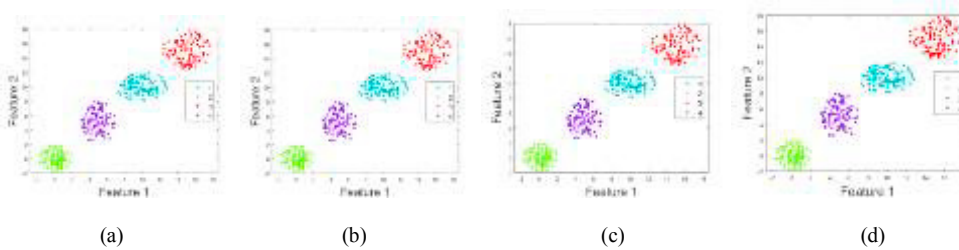


Fig. 9: Clustering results for AD_4_3 dataset using ADK-Means technique at different nodes : (a) node 1, (b) node 2, (c) node 3, (d) node 4

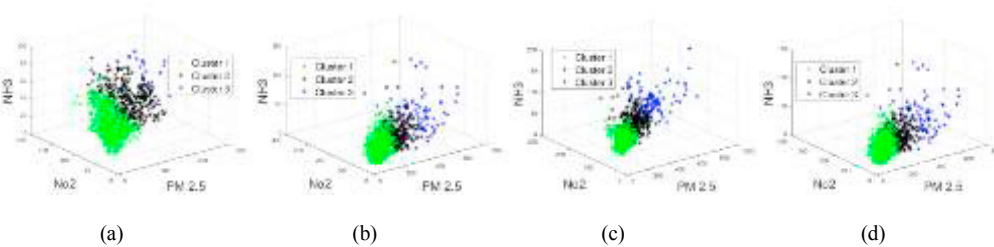


Fig. 10: Clustering results for Delhi air pollution dataset using ADK-Means technique at different nodes : (a) node 1, (b) node 2, (c) node 3, (d) node 4

5.2. Real life air pollution dataset results

This practical dataset has been obtained from central control room for air quality management, Delhi NCR, India. Central control room has installed many sensors at different stations to access the pollution content in air [7]. The data of four stations Alipur, Bawana, Dilshad garden and Jahagirpuri of Delhi, India has been considered for distributed clustering. Five parameters $PM_{2.5}$ ($\mu g/m^3$), NO_2 ($\mu g/m^3$), NH_3 ($\mu g/m^3$), SO_2 ($\mu g/m^3$), CO (mg/m^3) has been selected

for clustering. The hourly dataset recorded in March 2019 is considered for validation of clustering algorithm. The results of each sensor node using proposed method is shown in Fig.10

Table 1: Initial clusters taken (Z_{max}) and final clusters (Z) and comparative analysis of Minkoski Score (MS_{dc}), Silhouette (SI_{dc}) and Dunn index (DI_{dc}) for all datasets.

Index	Clusters		MS_{dc}		SI_{dc}		DI_{dc}	
Dataset	Z_{max}	Z	ADK-Means	DK-Means	ADK-Means	DK-Means	ADK-Means	DK-Means
Mixed_3_2	25	3	0.1766	0.2427	0.7874	0.8012	0.0859	0.0649
Mixed_5_2	30	5	0.1940	0.3662	0.7466	0.7600	0.0152	0.0143
AD_5_2	16	5	0.1894	0.5123	0.7213	0.6401	0.0412	0.0351
AD_4_3	20	4	0	0	0.8874	0.8874	0.6482	0.6482
Air pollution	28	3	-	-	0.7123	0.6201	0.0211	0.0162

6. Conclusion

In this paper an automatic distributed data clustering approach is developed which automatically selects number of clusters. The algorithm uses point symmetry based cluster assignment due to which it can detect arbitrary shape clusters. The algorithm does not require central base station. The distributed nature of algorithm makes uniform uses of power among the nodes thus improves the life time of sensor networks. The performance of the proposed algorithm is validated using cluster quality evaluation methods Minkowski score, Silhouette index and Dunn index. Four artificial dataset and one practical data set is used for simulation studies which gives better performance in comparison to existing DK-means clustering approach.

Acknowledgements

This work is supported by Ministry of Electronics and Information Technology (Miety) Gov. of India. Miety has provided Ph.D fellowship to Mr. Dinesh Kumar Kotarya to carry out research work at MNIT Jaipur, India.

References

- [1] Altilio, R., Di Lorenzo, P., Panella, M., 2019. Distributed data clustering over networks. Pattern Recognition 93, 603–620.
- [2] Ayyildiz, C., Erdem, H.E., Dirikgil, T., Dugenci, O., Kocak, T., Altun, F., Gungor, V.C., 2019. Structure health monitoring using wireless sensor networks on structural elements. Ad Hoc Networks 82, 68–76.

- [3] Azimi, R., Sajedi, H., 2018. A decentralized gossip based approach for data clustering in peer-to-peer networks. *Journal of Parallel and Distributed Computing* 119, 64–80.
- [4] Azimi, R., Sajedi, H., Ghayekhloo, M., 2017. A distributed data clustering algorithm in p2p networks. *Applied Soft Computing* 51, 147–167.
- [5] Bandyopadhyay, S., Giannella, C., Maulik, U., Kargupta, H., Liu, K., Datta, S., 2006. Clustering distributed data streams in peer-to-peer environments. *Inf. Sci.* 176, 1952–1985.
- [6] Bandyopadhyay, S., Saha, S., 2007. Gaps: A clustering method using a new point symmetry-based distance measure. *Pattern recognition* 40, 3430–3451.
- [7] CCR, 2019. Central control room for air quality management. URL: <https://app.cpcbccr.com/ccr/#/login>.
- [8] Datta, Souptik, G.C., Kargupta, H., 2009. Approximate distributed k-means clustering over a peer-to-peer network. *IEEE Transactions on Knowledge and Data Engineering* 21, 1372–1388.
- [9] Forero, P.A., Cano, A., Giannakis, G.B., 2008. Consensus-based k-means algorithm for distributed learning using wireless sensor networks, in: *Proceedings of the Workshop on Sensors, Signal and Info. Process.*, Sedona, AZ, pp. 11–14.
- [10] Nanda, S.J., Panda, G., 2014. A survey on nature inspired metaheuristic algorithms for partitional clustering. *Swarm and Evolutionary computation* 16, 1–18.
- [11] Pedrycz, W., Rai, P., 2008. Collaborative clustering with the use of fuzzy c-means and its quantification. *Fuzzy Sets and Systems* 159, 2399–2427.
- [12] Saha, S., Bandyopadhyay, S., 2009. A new multiobjective simulated annealing based clustering technique using symmetry. *Pattern Recognition Letters* 30, 1392–1403.
- [13] Su, M.C., Chou, C.H., 2001. A modified version of the k-means algorithm with a distance based on cluster symmetry. *IEEE Transactions on Pattern Analysis & Machine Intelligence*, 674–680.